UAS Avionics Enclosure

Structural & Thermal Qualification — Technical Report

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Project	Virtual environmental qualification of a machined avionics enclosure
Platform	7 in class long-range multirotor UAS, 6S LiPo
Material	6061-T6 aluminium, CNC machined
Tools	Fusion 360 (Nastran modal FEA), Python, Excel
Revision	C — modal results incorporated
Status	Structural analysis complete. Thermal redesign and random vibration open.

1. Scope

Defines the design requirements and environmental load cases for an avionics enclosure housing the flight control, propulsion control and communications stack of a 7 in class multirotor UAS.

Verification is by analysis only; no physical test article is produced. Every requirement traces to a manufacturer-published limit, a published military standard, or a platform-derived excitation condition.

- **In scope:** structural dynamics (modal, random vibration), thermal (conduction, free and forced convection), mass, envelope.
- **Out of scope:** shock, EMI/EMC, ingress protection, altitude, humidity, physical test.

2. Component stack

Baseline selected by weighted trade study across six categories, scored on the parameters that drive enclosure design — dissipation, mass, envelope and temperature limit — rather than flight performance.

ID	Component	Part	Mass (g)	Footprint (mm)	Height (mm)	Diss. (W)	Location
C1	Flight controller	Matek H743-SLIM	7.0	36 × 36	5.0	2.00	Internal
C2	4-in-1 ESC	Holybro Tekko32 F4 65A	15.8	46 × 44	6.0	0.90	Internal
C3	Video transmitter	RushFPV Tank Ultimate	12.0	37 × 24	6.7	**7.90**	Internal
C4	GPS / compass	Matek M10Q-5883	8.0	20 × 20	12.4	0.07	External
C5	RC receiver	Happymodel EP2 ELRS	0.4	10 × 10	6.0	0.20	External
C6	Bulk capacitor	35PK1000MEFCT810X20	2.75	⌀ 11.25	25.0	negligible	Internal
Internal heat load (C1 + C2 + C3)						**10.80**	**—**

> **C4 and C5 are mounted externally.** An aluminium enclosure forms a Faraday cage. The M10Q-5883 uses an internal patch antenna and the EP2 an integrated SMD ceramic antenna; neither would acquire signal if enclosed. Two thermal requirements were deleted as a direct consequence.

Temperature limits. Two distinct limits apply and must not be conflated. C1 carries a 40 °C maximum *ambient* rating alongside a 100 °C component rating; inside a sealed enclosure the ambient limit governs internal air. The binding internal *component* allowable is 90 °C, set by C3.

C2 has no output BEC, so all 5 V loads draw from C1 and the conversion losses dissipate there — which is why C1 is booked at 2.00 W rather than its published 1.0 W static figure. C2's larger 46 × 44 mm outline sets the minimum internal footprint.

3. Environments

The propulsion system does not sit inside the enclosure. It is specified because it defines the vibration environment. Prop passing frequency is $f = (RPM / 60) \times N_{blades}$.

Parameter	Value	Harmonic	At idle (2,600 RPM)	At max (17,200 RPM)
Motor	T-Motor F90 2806.5, 1500 KV	1×	87 Hz	573 Hz
Propeller	7 in, 2 blades	2×	173 Hz	1,147 Hz
RPM range	2,600 – 17,200	3×	260 Hz	1,720 Hz
Source	Manufacturer thrust table	4×	347 Hz	**2,293 Hz**

> **Separation factor.** 1.5× on the fourth harmonic at maximum RPM gives the 3,440 Hz target — a common preliminary-design rule for avionics mounting structure. A 2.0× factor would raise the target to 4,587 Hz and **the design would not close**. Recorded as a sensitivity rather than a failure, because the bonded-contact assumption (A6) already over-predicts stiffness.

Thermal conditions. Hover: ≈0 m/s freestream (prop wash only), 35 °C ambient — worst-case convection, lowest ESC load. Cruise: 18 m/s, 35 °C ambient — better convection, higher I^2R loss. The two cases trade against each other, so neither is binding a priori.

4. Requirements

ID	Requirement	Source	Verification
REQ-STR-001	First natural frequency shall exceed 3,440 Hz	1.5× separation on 4× harmonic	Modal FEA
REQ-STR-002	No mode shall fall within 87 to 2,293 Hz	Excitation band	Modal FEA
REQ-STR-003	MS under 3σ random vibration shall be positive against 6061-T6 yield	MIL-STD-810H	Miles' equation + static FEA
REQ-STR-004	Mounting interface shall survive cumulative exposure without fatigue failure	6061-T6 S-N data	Fatigue hand calc
REQ-THM-001	Internal air shall not exceed 40 °C in any flight condition	C1 max ambient	Thermal FEA
REQ-THM-002	C1 component temperature shall not exceed 100 °C	C1 spec	Thermal FEA
REQ-THM-003	C2 MOSFET temperature shall not exceed 150 °C	C2 spec	Thermal FEA
REQ-THM-004	C3 case temperature shall not exceed 90 °C	C3 spec	Thermal FEA
REQ-THM-005	C6 capacitor temperature shall not exceed its rating	C6 spec	Thermal FEA
REQ-MAS-001	Enclosure mass shall not exceed the platform allocation	Mass budget	CAD mass properties
REQ-ENV-001	Internal volume shall accommodate C1, C2, C3, C6 with clearance	§2	CAD
REQ-ENV-002	Enclosure shall accept the 30.5 × 30.5 mm stack mounting pattern	C1 / C2 interface	CAD
REQ-ENV-003	Enclosure shall provide a cable penetration for the external C4 / C5 harness	RF integration	CAD

5. Margin table

Req ID	Condition	Predicted	Allowable	Margin	Status
REQ-STR-001	Modal	3,984 Hz	3,440 Hz	**+0.16**	**PASS**
REQ-STR-002	Modal	f_1 = 3,984 Hz	87–2,293 Hz	**No mode in band**	**PASS**
REQ-STR-003	Random vibration	—	—	—	**OPEN**
REQ-STR-004	Fatigue	—	—	—	**OPEN**
REQ-THM-001	Hover, bare Al	**86.0 °C**	40 °C	**−0.53**	**FAIL**
REQ-THM-001	Hover, anodised	**69.4 °C**	40 °C	**−0.42**	**FAIL**
REQ-THM-001	Cruise, bare Al	**43.9 °C**	40 °C	**−0.09**	**FAIL**
REQ-THM-001	Cruise, anodised	**43.2 °C**	40 °C	**−0.07**	**FAIL**
REQ-THM-002/3/4	Component temperatures	—	100 / 150 / 90 °C	—	**OPEN**
REQ-MAS-001	Base mass	96 g	—	—	**OPEN**

MS = (Allowable / Predicted) − 1. Thermal predictions come from the preliminary lumped-parameter calculation, not FEA. They are surface temperatures and therefore optimistic; internal air will run hotter.

6. Modal FEA

Ten modes extracted in Fusion 360 (Nastran). Constraint: fixed on the four mounting tab underside faces. Contacts: bonded at all interfaces (A6). **No mode falls within the excitation band.**

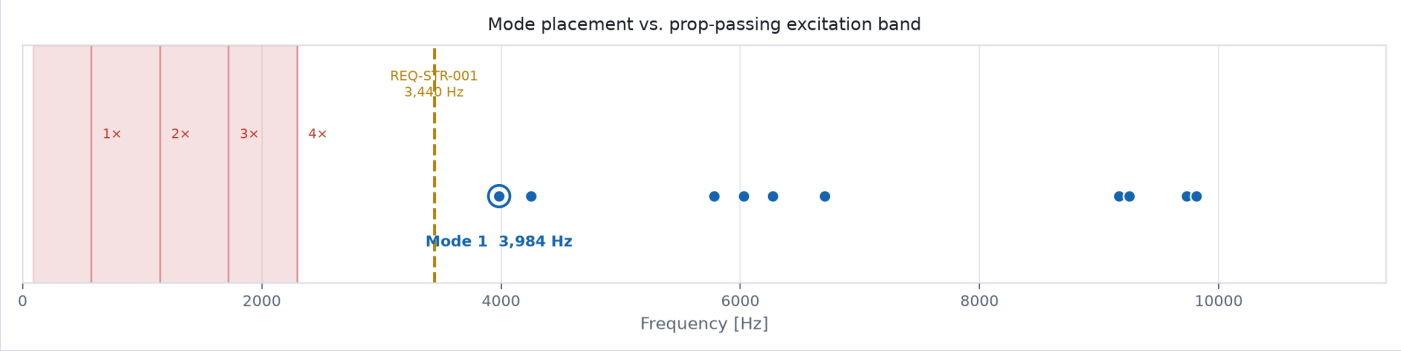


Figure 1 — All ten modes against the prop-passing excitation band. The first mode at 3,984 Hz clears the 3,440 Hz requirement and the fourth harmonic at maximum RPM by a factor of 1.74.

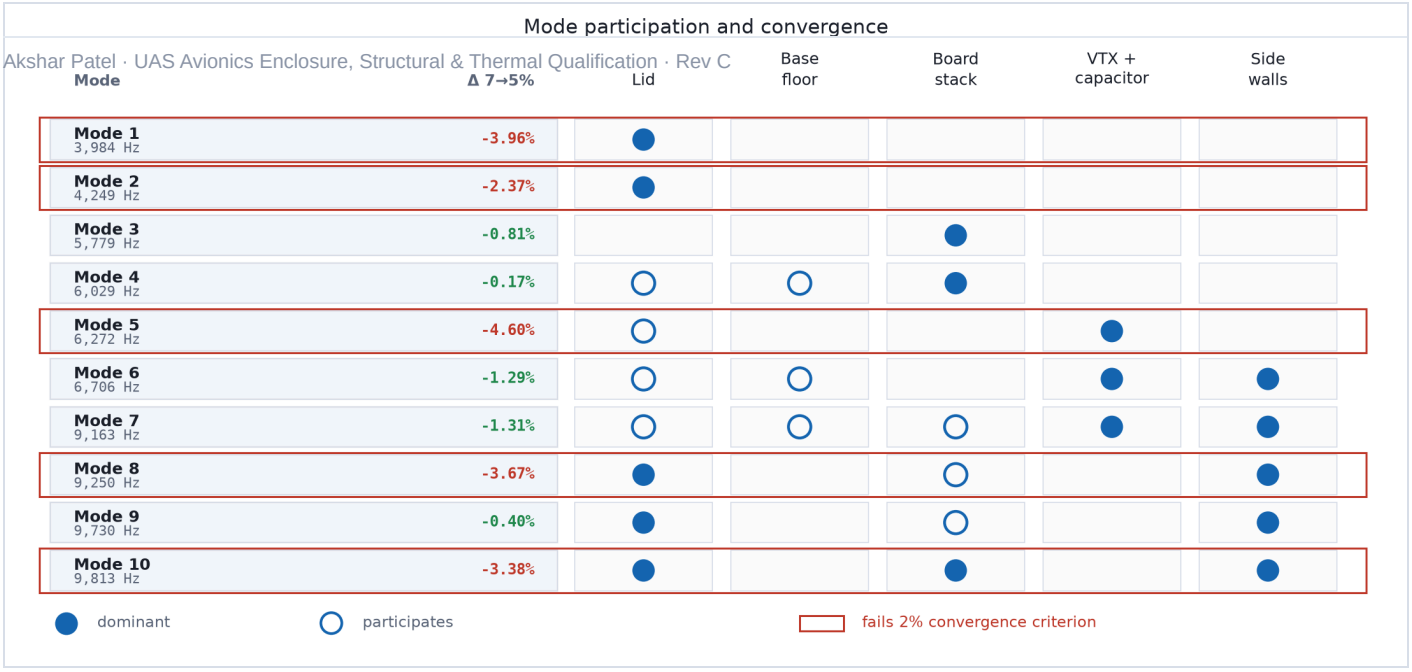


Figure 2 — Which parts of the enclosure move in each mode, with convergence status. Every mode that fails the 2% criterion has the lid or the capacitor as a dominant participant — the two parts sitting on bonded contact between separately meshed bodies.

Mode	Freq (5% mesh)	Shape
1	3,984 Hz	Lid drumming, first bending. Centre antinode decaying to zero at the four corner bolts; base and internal stack essentially rigid.
2	4,249 Hz	Second lid mode. Lid deforms, base floor does not participate.
3	5,779 Hz	Board stack responds about the Y axis. VTX and capacitor stationary; no lid or floor bending.
4	6,029 Hz	Small coupled lid and floor bending. Outer walls stationary; board stack bends about X.
5	6,272 Hz	Capacitor and partial VTX response. Lid deforms out of plane over the stack, opposite half inward.
6	6,706 Hz	Capacitor-side wall deforms inward with the VTX. Lid and floor deform in opposition across the halves.
7	9,163 Hz	Capacitor deformation coupled into the adjacent wall. Lid and floor in anti-phase diagonal pattern.
8	9,250 Hz	Coupled lid and long-wall response. Stack diagonal pattern peaking at one ESC corner; capacitor near-stationary.
9	9,730 Hz	Long-wall out-of-plane bending. Two lid antinodes either side of the cable penetration; base floor and tabs nodal.
10	9,813 Hz	Higher-order side-wall mode, three antinodes along the long wall. Lid bows up while the stack bows down; standoffs couple the two.

7. Mesh convergence

Three meshes at 10%, 7% and 5% element size. A shift below 2% is taken as converged.

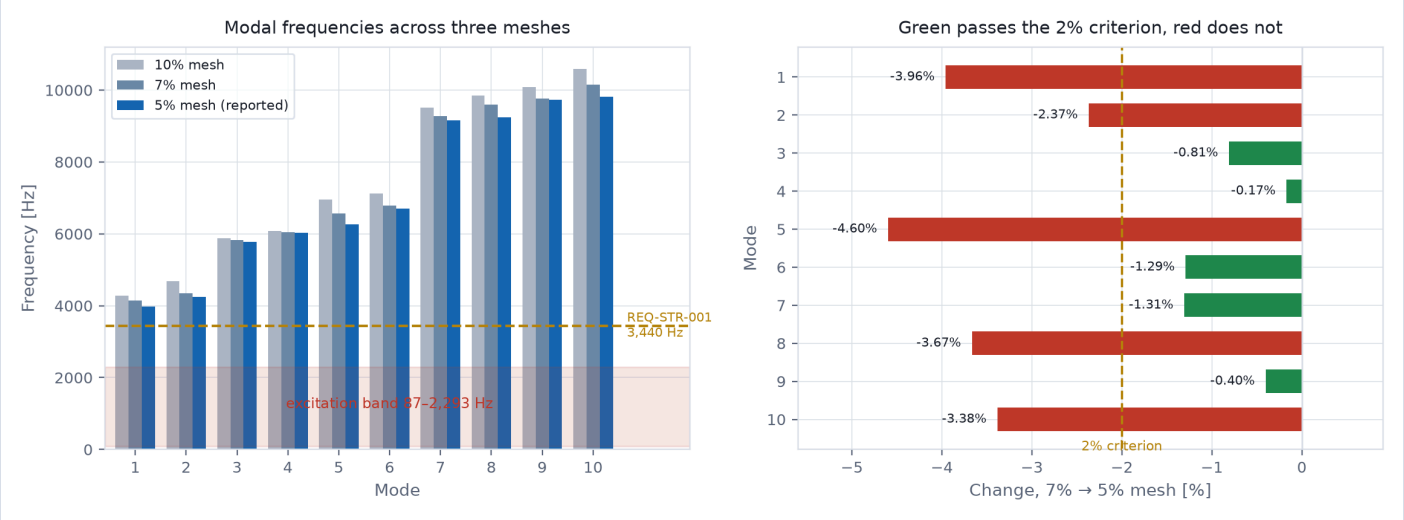


Figure 3 — Modal frequencies across all three meshes, and per-mode convergence against the 2% criterion.

Mode	10% (baseline)	7% (refined)	Δ 10→7	5% (further refined)	Δ 7→5
1	4,286.5	4,147.9	−3.23%	3,983.7	−3.96%
2	4,690.0	4,352.5	−7.20%	4,249.4	−2.37%
3	5,882.3	5,826.4	−0.95%	5,779.2	−0.81%
4	6,071.6	6,038.9	−0.54%	6,028.6	−0.17%
5	6,955.8	6,574.2	−5.49%	6,271.9	−4.60%
6	7,130.9	6,793.3	−4.73%	6,705.6	−1.29%
7	9,520.8	9,284.6	−2.48%	9,163.3	−1.31%
8	9,847.4	9,601.5	−2.50%	9,249.5	−3.67%
9	10,081.1	9,769.5	−3.09%	9,730.2	−0.40%
10	10,591.3	10,156.3	−4.11%	9,812.7	−3.38%

Mode 1 does not converge. Its shift grew from 3.23% to 3.96% across successive refinements rather than shrinking, and Richardson extrapolation returns a negative observed order of convergence — the discretisation error is not reducing monotonically. Modes 3 and 4 converge cleanly at 0.81% and 0.17%, and both are dominated by the base, a single solid body with no contact dependence. Every poorly converging mode (1, 2, 5, 8, 10) involves the lid or the capacitor, which depend on bonded contact between separately meshed bodies. Contact discretisation at those interfaces is the probable cause, not global mesh density. The reported 3,983.7 Hz is the lowest of the three meshes and therefore the most conservative available estimate.

8. Thermal scoping

Lumped-parameter hand calculation in Python, built before any FEA to establish an expected answer to check the FEA against. Churchill-Chu natural convection, flat-plate laminar forced convection and linearised radiation, solved by fixed-point iteration for a single isothermal surface temperature.

> **Limits of this calculation.** It does not resolve internal gradients, component hot spots or conduction through mounts — those require thermal FEA, which has not been run. No CFD has been performed on this project. Because the enclosure is treated as isothermal, internal air will run hotter than these figures: treat them as optimistic.

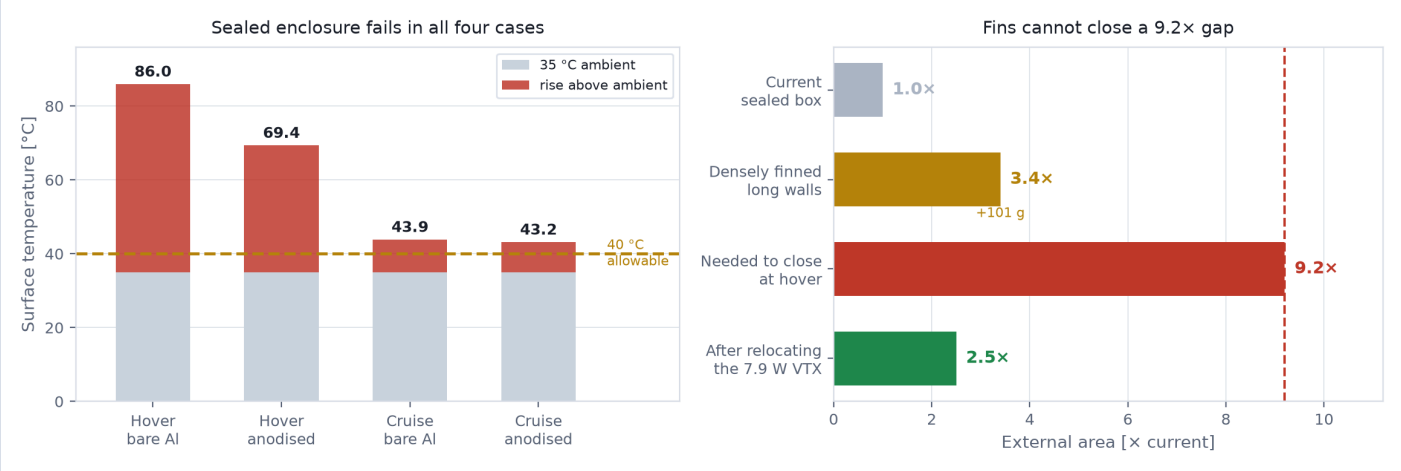


Figure 4 — Left: surface temperature for all four cases against the 40 °C allowable. Right: external area required to close the gap at hover.

Case	h_{conv}	h_{rad}	ΔT	$T_{surface}$	Margin	Status
Hover, bare Al	8.7	0.8	51.0 °C	**86.0 °C**	**−0.53**	**FAIL**
Hover, anodised	8.0	6.3	34.4 °C	**69.4 °C**	**−0.42**	**FAIL**
Cruise, bare Al	54.6	0.7	8.9 °C	**43.9 °C**	**−0.09**	**FAIL**
Cruise, anodised	54.6	5.5	8.2 °C	**43.2 °C**	**−0.07**	**FAIL**

h in W/m²·K. External area 0.0221 m². Cruise Reynolds number 93,095 — laminar, so the flat-plate laminar correlation is appropriate.

Hover is binding: worst heat rejection despite the lowest ESC load. Anodising buys 16.6 °C at hover because radiation carries 6.3 of the 14.3 W/m²·K total against 0.8 bare, making surface finish a first-order variable. Closing a 34 °C rise against a 5 °C allowable requires **9.2× the external area**. Fins were sized rather than assumed: densely finned long walls reach only 3.4× while adding 101 g, more than doubling the 96 g base. Relocating the 7.9 W video transmitter externally removes 73% of the internal load and drops the requirement to 2.5×, which fins can deliver.

9. Assumptions log

#	Assumption	Justification	Impact if wrong
A1	C1 dissipation taken as 2.00 W	Published static power is 1.0 W; rounded up to bound BEC conversion losses	**Over-predicts heat load by ~0.6 W. Conservative.**
A2	C2 dissipation 0.90 W at hover	Conduction loss scales as $I^2 \cdot R_{ds(on)}$	**A 2× current error is a 4× dissipation error**
A3	Ambient 35 °C	Hot-day condition for the intended environment	**Directly offsets every predicted temperature**
A4	C1 40 °C read as max ambient	Distinct from the ~100 °C component rating; in a sealed box the ambient limit governs	**If it is a component limit, REQ-THM-001 is over-conservative**
A5	Standoffs modelled as aluminium	Stiffer than the nylon common in FPV builds	**Nylon would lower board stack modes. NON-CONSERVATIVE.**
A6	All interfaces bonded	Represents fully preloaded bolted joints with no slip	**Over-predicts stiffness and natural frequencies. NON-CONSERVATIVE for REQ-STR-001.**
A7	Components as homogeneous solids	Correct mass and envelope; FEA needs mass distribution, not board layout	**Local board stiffness unrepresented**

> **Two assumptions work against the design, and both touch the one requirement that passes.** Bonded contacts (A6) over-predict stiffness, and aluminium standoffs (A5) are stiffer than the nylon a real build would likely use. Both inflate natural frequencies — the direction that makes REQ-STR-001 easier to satisfy. The +0.16 margin is an upper bound on the true margin.

10. Open items

- **Phase 5 thermal redesign** — relocate the VTX, fin to the 2.5× target, re-run. Decides whether the design closes.
- **Thermal FEA** — resolve internal gradients and hot spots. A disagreement with the hand calculation beyond ~30% needs explaining before either result is trusted.
- **Random vibration and fatigue** — Miles' equation for 3σ response, static FEA against 6061-T6 yield, fatigue hand calc on the mounting interface.
- **MIL-STD-810H Method 514.8** — obtain and extract the PSD profile with a stated rationale for a small multirotor.
- **Datasheet gaps** — confirm the C1 40 °C basis, state assumed ESC phase current for both conditions, confirm C3 runs from VBAT, obtain the C6 temperature rating.
- **Mass budget** — REQ-MAS-001 has no allocation. Base is 96 g; a finned redesign adds materially.